IEC 61508 Assurance-as-Code Demo documentation

This site demonstrates a minimal, end-to-end IEC 61508 safety lifecycle thread for a SIL 2 Emergency Stop, encoded as structured, traceable engineering artifacts using Sphinx and Sphinx-Needs.

Overview

This demo encodes a minimal IEC 61508 safety lifecycle thread for a SIL 2 Emergency Stop as structured, typed engineering artifacts using Sphinx and Sphinx-Needs.

The example spans:

- Hazard and risk assessment
- SIL allocation
- Functional and technical safety requirements
- Safety architecture
- Verification activities
- Evidence artifacts
- Confirmation measures

Unlike conventional document-centric safety documentation, each lifecycle element is represented as a typed object with explicitly declared relations. Allowed artifact types and link semantics are constrained by a predefined meta-model and validated at build time.

Traceability graphs and lifecycle completeness checks are computed directly from these declared relations. Missing structural links (e.g., requirement without verification) are automatically detected.

Scope

This demonstration focuses strictly on structural lifecycle consistency.

It does not evaluate:

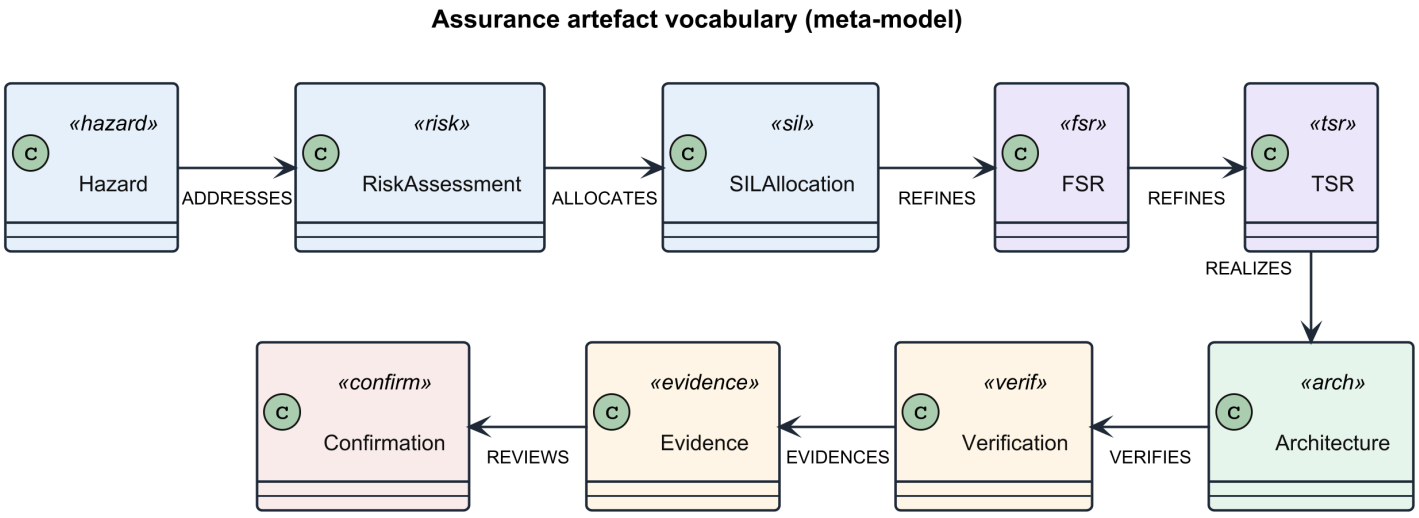
- Semantic correctness of hazard analysis
- Quantitative SIL determination
- Architectural metrics (e.g., HFT, SFF)

- Full safety case argumentation

The purpose is to demonstrate executable lifecycle structure and automated traceability enforcement for a single SIL 2 thread.

Meta-model

The figure below defines the assurance artifact vocabulary used in the demo.

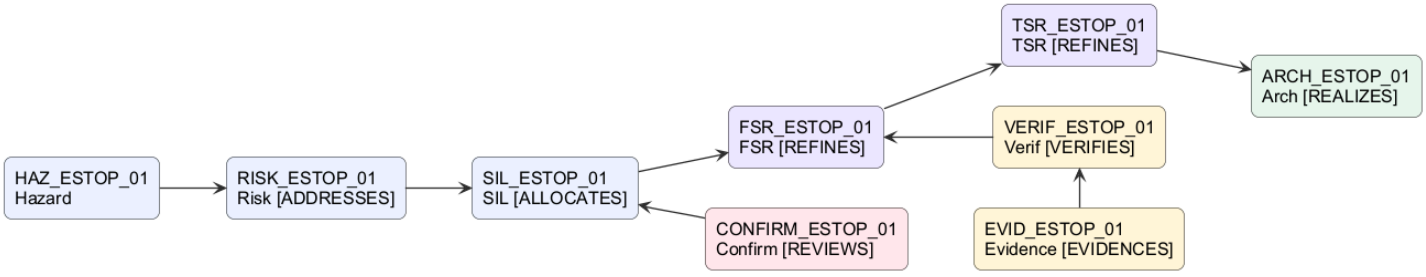


Assurance artefact meta-model (demo vocabulary). This vocabulary is enforced during build; only these link types are used in computed trace graphs and completeness checks.

Workflow

This section describes the intended IEC 61508 lifecycle flow for the SIL 2 Emergency Stop thread.

The figure below illustrates the workflow, showing the key assurance artifacts and their relationships.



IEC 61508 SIL 2 Emergency Stop assurance workflow. Each node corresponds to a Sphinx-Needs artefact ID; each edge label corresponds to a validated link type used to generate traceability views and checks.

1. Hazard

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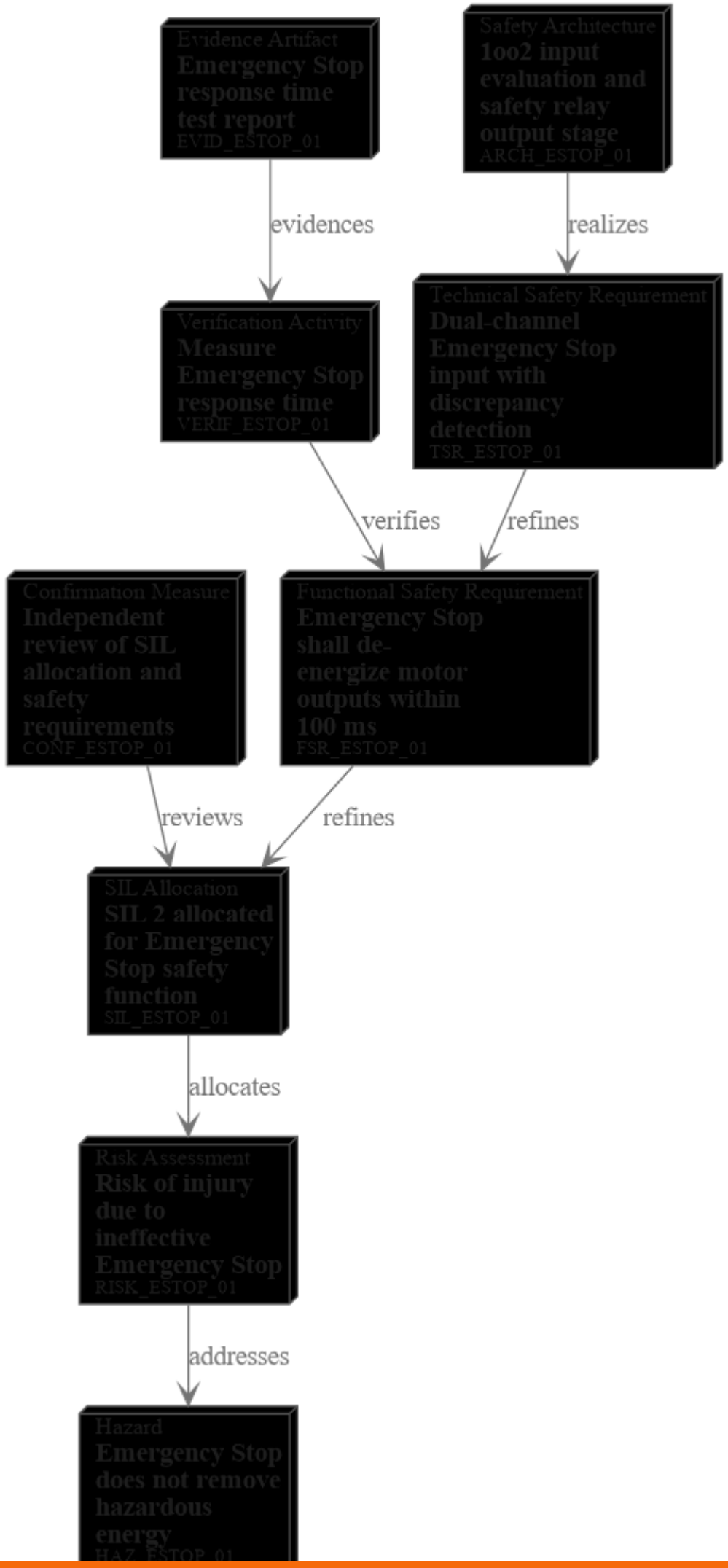
3. SIL allocation
4. Functional safety requirements (FSR)
5. Technical safety requirements (TSR)
6. Safety architecture
7. Verification activities
8. Evidence artifacts
9. Confirmation measures

Each node corresponds to a Sphinx-Needs artefact ID; each edge label corresponds to a validated link type used to generate traceability views and checks.

Assurance Graph



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Lifecycle Artifacts

This section encodes a minimal IEC 61508 SIL 2 lifecycle thread for an Emergency Stop function as structured Sphinx-Needs artifacts.

Hazard

Hazard: **Emergency Stop does not remove hazardous energy** HAZ_ESTOP_01

phase: Hazard and Risk Analysis
owner_role: Safety Engineer
part_ref: IEC 61508-1 (Hazard and Risk Analysis)
addressed_by: RISK_ESTOP_01 [Risk Assessment]

If the Emergency Stop request is not effective, the machine may continue hazardous motion, exposing personnel to injury.

Risk Assessment

Risk Assessment: **Risk of injury due to ineffective Emergency Stop** RISK_ESTOP_01

phase: Hazard and Risk Analysis
owner_role: Safety Engineer
part_ref: IEC 61508-1 (Risk Assessment)
addresses: HAZ_ESTOP_01 [Hazard]
allocated_by: SIL_ESTOP_01 [SIL Allocation]

Risk evaluation indicates that risk reduction is required for the Emergency Stop function.

SIL Allocation

SIL Allocation: **SIL 2 allocated for Emergency Stop safety function** SIL_ESTOP_01

phase: Safety Function Allocation
sil_level: SIL2
owner_role: Functional Safety Manager
part_ref: IEC 61508-1 (SIL Allocation)

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reviewed_by: CONF_ESTOP_01 [Confirmation Measure]

Based on the risk assessment, the Emergency Stop safety function is allocated SIL 2.

Functional Safety Requirement

Functional Safety Requirement: **Emergency Stop shall de-energize motor outputs within 100 ms** FSR_ESTOP_01

phase: Safety Requirements Specification
sil_level: SIL2
owner_role: System Engineer
part_ref: IEC 61508-2 (SRS)
refines: SIL_ESTOP_01 [SIL Allocation]
refined_by: TSR_ESTOP_01 [Technical Safety Requirement]
verified_by: VERIF_ESTOP_01 [Verification Activity]

When the Emergency Stop is activated, the system shall remove power from motor outputs within 100 ms.

Technical Safety Requirement

Technical Safety Requirement: **Dual-channel Emergency Stop input with discrepancy detection** TSR_ESTOP_01

phase: Safety Requirements Specification
sil_level: SIL2
owner_role: Hardware Engineer
part_ref: IEC 61508-2 (SRS)
refines: FSR_ESTOP_01 [Functional Safety Requirement]
realized_by: ARCH_ESTOP_01 [Safety Architecture]

The Emergency Stop shall be implemented with two independent input channels and discrepancy detection to detect wiring or contact faults.

Safety Architecture

Safety Architecture: **1oo2 input evaluation and safety relay output stage** ARCH_ESTOP_01

phase: Architectural Design
owner_role: System Architect
part_ref: IEC 61508-2 (Architecture)
realizes: TSR_ESTOP_01 [Technical Safety Requirement]

The architecture evaluates both input channels and commands a safety relay / safe output stage to remove motor power.

Verification Activity

Verification Activity: **Measure Emergency Stop response time** VERIF_ESTOP_01

phase: Verification
owner_role: Test Engineer
part_ref: IEC 61508-3 (Verification)
verifies: FSR_ESTOP_01 [Functional Safety Requirement]
evidenced_by: EVID_ESTOP_01 [Evidence Artifact]

Evidence

Evidence Artifact: **Emergency Stop response time test report** EVID_ESTOP_01

phase: Verification Evidence
owner_role: Test Engineer
artifact_kind: Test Report
evidences: VERIF_ESTOP_01 [Verification Activity]

Evidence artifact documenting the measured response time and acceptance criteria.

Confirmation Measure

Confirmation Measure: **Independent review of SIL allocation and safety requirements** CONF_ESTOP_01

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phase: Functional Safety Assessment
owner_role: Independent Assessor
part_ref: IEC 61508-1 (Confirmation Measures)
reviews: SIL_ESTOP_01 [SIL Allocation]

Confirmation activity records independent review of the SIL allocation decision and the derived safety requirements.

Traceability Views

This section provides traceability views over the SIL 2 Emergency Stop lifecycle artifacts.

All lifecycle artifacts

Show 10 entries Columns Copy Excel PDF Search:

ID	Title	Type	Phase	Sil Level	Owner Role
ARCH_ESTOP_01	1oo2 input evaluation and safety relay output stage	arch	Architectural Design		System Architect
CONF_ESTOP_01	Independent review of SIL allocation and safety requirements	confirm	Functional Safety Assessment		Independent Assessor
EVID_ESTOP_01	Emergency Stop response time test report	evidence	Verification Evidence		Test Engineer
FSR_ESTOP_01	Emergency Stop shall de-energize motor outputs within 100 ms	fsr	Safety Requirements Specification	SIL2	System Engineer
HAZ_ESTOP_01	Emergency Stop does not remove hazardous energy	hazard	Hazard and Risk Analysis		Safety Engineer
RISK_ESTOP_01	Risk of injury due to ineffective Emergency	risk	Hazard and Risk Analysis		Safety Engineer

SIL_ESTOP_01	SIL 2 allocated for Emergency Stop safety function	sil	Safety Function Allocation	SIL2	Functional Safety Manager
TSR_ESTOP_01	Dual-channel Emergency Stop input with discrepancy detection	tsr	Safety Requirements Specification	SIL2	Hardware Engineer
VERIF_ESTOP_01	Measure Emergency Stop response time	verif	Verification		Test Engineer

Showing 1 to 9 of 9 entries

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Emergency Stop thread (filtered by ID prefix)

Show 10 entries Columns Copy Excel PDF Search:

ID	Title	Type	Phase	Sil Level
ARCH_ESTOP_01	1oo2 input evaluation and safety relay output stage	arch	Architectural Design	
CONF_ESTOP_01	Independent review of SIL allocation and safety requirements	confirm	Functional Safety Assessment	
EVID_ESTOP_01	Emergency Stop response time test report	evidence	Verification Evidence	
FSR_ESTOP_01	Emergency Stop shall de-energize motor outputs within 100 ms	fsr	Safety Requirements Specification	SIL2
HAZ_ESTOP_01	Emergency Stop does not remove hazardous energy	hazard	Hazard and Risk Analysis	
RISK_ESTOP_01	Risk of injury due to ineffective Emergency Stop	risk	Hazard and Risk Analysis	
SIL_ESTOP_01	SIL 2 allocated for Emergency Stop safety function	sil	Safety Function Allocation	SIL2

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TSR_ESTOP_01	Dual-channel Emergency Stop input with discrepancy detection	tsr	Safety Requirements Specification	SIL2
VERIF_ESTOP_01	Measure Emergency Stop response time	verif	Verification	

Showing 1 to 9 of 9 entries

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Verification coverage (FSR without verification)

The following table lists functional safety requirements that do not yet have a verification activity linked.

Show 10 entries Columns Copy Excel PDF Search:

ID	Title	Phase	Sil Level	Owner Role
FSR_ESTOP_01	Emergency Stop shall de-energize motor outputs within 100 ms	Safety Requirements Specification	SIL2	System Engineer

Showing 1 to 1 of 1 entries

Previous 1 Next

Evidence coverage (verification without evidence)

The following table lists verification activities that do not yet have evidence linked.

Show 10 entries Columns Copy Excel PDF Search:

ID	Title	Phase	Owner Role
VERIF_ESTOP_01	Measure Emergency Stop response time	Verification	Test Engineer

Showing 1 to 1 of 1 entries

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SIL allocation coverage (risk without SIL allocation)

The following table lists risk assessment items that do not yet have a SIL allocation linked.

Show 10 entries Columns Copy Excel PDF Search:

ID	Title	Phase	Owner Role
RISK_ESTOP_01	Risk of injury due to ineffective Emergency Stop	Hazard and Risk Analysis	Safety Engineer

Showing 1 to 1 of 1 entries

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Lifecycle Checks

This section defines simple, automated completeness checks for the SIL 2 Emergency Stop thread. The goal is to detect missing lifecycle links early (e.g., requirements without verification).

Check 1: Every FSR must be verified

Expected rule: Each Functional Safety Requirement (FSR) shall have at least one incoming `verifies` link from a verification activity.

Show 10 entries Columns Copy Excel PDF Search:

ID	Title	Sil Level	Owner Role
FSR_ESTOP_01	Emergency Stop shall de-energize motor outputs within 100 ms	SIL2	System Engineer

Showing 1 to 1 of 1 entries

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If this table is non-empty, verification coverage is incomplete.

Check 2: Every verification must have evidence

Expected rule: Each verification activity shall have at least one incoming `evidences` link from an evidence artifact.

Show 10 entries Columns Copy Excel PDF Search:

By Dr. Padma Iyenghar

ID	Title	Owner Role
VERIF_ESTOP_01	Measure Emergency Stop response time	Test Engineer

Showing 1 to 1 of 1 entries

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Check 3: Every risk item must have a SIL allocation

Expected rule: Each risk assessment entry shall have at least one incoming `allocates` link from a SIL allocation artifact.

Show 10 entries Columns Copy Excel PDF Search:

ID	Title	Owner Role
RISK_ESTOP_01	Risk of injury due to ineffective Emergency Stop	Safety Engineer

Showing 1 to 1 of 1 entries Previous 1 Next

If this table is non-empty, SIL allocation coverage is incomplete.

Check 4: Every hazard must be addressed by a risk item

Expected rule: Each hazard shall be addressed by at least one risk assessment entry.

Show 10 entries Columns Copy Excel PDF Search:

ID	Title	Owner Role
HAZ_ESTOP_01	Emergency Stop does not remove hazardous energy	Safety Engineer

Showing 1 to 1 of 1 entries Previous 1 Next

If this table is non-empty, some hazards are not covered by any risk assessment.